

Comprehensive IPv6 Assignment Guidelines for CSE Undergraduates

1. Introduction to IPv6

IPv6 addresses critical limitations of IPv4, primarily **exhaustion of 32-bit addresses**, by introducing a **128-bit address space** (340 undecillion addresses). Its adoption exceeds 50% globally due to improved routing efficiency, built-in security (IPsec), and simplified network configuration. Understanding IPv6 is essential for modern network design, IoT implementations, and cloud infrastructure .

2. Core IPv6 Topics & Learning Objectives

a) Address Architecture

Structure: 128-bit address, written in hexadecimal (e.g., 2001:db8:85a3::8a2e:0370:7334), grouped into eight 16-bit blocks separated by colons.

Shortening Rules: Leading zeros omitted; consecutive zero blocks replaced by :: (e.g., 2001:0db8::0001 → 2001:db8::1) .

Prefix Notation: Network portion denoted by / (e.g., /64).

b) Address Types

Type	Prefix	Scope	Use Case
Global Unicast	2000::/3	Internet-wide	Public-facing servers
Unique Local (ULA)	fd00::/8	Internal networks	Private infrastructures
Link-Local	fe80::/10	Single network segment	Router discovery, neighbor comms
Multicast	ff00::/8	Defined by scope	Group communication (e.g., video streaming)
Anycast	From Unicast	Nearest interface	Load balancing, DNS services

c) Configuration Mechanisms

SLAAC (Stateless Address Autoconfiguration):

Hosts generate addresses using **Router Advertisements (RA)** + **EUI-64** (expands 48-bit MAC to 64-bit by inserting ff:fe and inverting the 7th bit) .

DHCPv6:

Stateful: Server assigns addresses (like IPv4 DHCP).

Stateless: Server provides DNS/domain info; hosts use SLAAC for addresses.

Configuration files (e.g., Linux: /etc/dhcp/dhcpd6.conf) define subnets and options .

d) Key Protocols

Neighbor Discovery Protocol (NDP): Replaces ARP; resolves MAC addresses via **Neighbor Solicitation/Advertisement** messages .

ICMPv6: Manages errors, diagnostics (e.g., ping6), and path MTU discovery .

e) Subnetting & Routing

Subnetting: A /64 prefix is standard for end-user networks. Example: Split 2001:db8:1::/48 into 65,536 /64 subnets .

Address Aggregation: Summarize contiguous blocks (e.g., 2001:db8:1::/48) to minimize routing tables .

f) Transition Mechanisms

Dual Stack: Simultaneous IPv4/IPv6 support.

Tunneling: Encapsulate IPv6 in IPv4 packets (e.g., 6to4) .

g) Security & Privacy

IPsec: Mandatory for encryption/authentication.

Privacy Extensions: Temporary random addresses to prevent tracking .

3. Practical Assignments

a) Basic Configuration (Week 1-2)

Task 1: Configure IPv6 on Ubuntu:

```
sudo ip -6 addr add 2001:db8::1/64 dev eth0
```

Task 2: Verify connectivity using ping6 and traceroute6 .

b) DHCPv6 Server Setup (Week 3-4)

Steps:

1. Install dhcpcd6 on Linux.
2. Edit /etc/dhcp/dhcpcd6.conf:

```
subnet6 2001:db8:1:2::/64 {  
  range6 2001:db8:1:2::100 2001:db8:1:2::200;  
  option dhcp6.name-servers 2001:db8:1:2::1;  
}
```

3. Enable SELinux/firewall rules .

c) Subnet Design (Week 5-6)

Scenario: Allocate a /48 prefix across three departments, each requiring 10 subnets. Calculate subnet IDs and host ranges.

d) AWS IPv6 Deployment (Week 7-8)

Tasks:

1. Deploy a VPC with **Amazon-provided GUA** (public) and **ULA** (private).
2. Configure security groups for IPv6 traffic .

e) Troubleshooting (Week 9)

Case Study: Diagnose SLAAC failures caused by misconfigured RA flags (e.g., ManagedAddressConfiguration bit) .

4. Assessment Rubric

Criteria	Weight	Description
Conceptual Accuracy	30%	Correct use of terminology (e.g., EUI-64, NDP, ULA)
Technical Implementation	40%	Error-free configuration (DHCPv6, subnetting)
Analysis	20%	Critical evaluation of design choices (e.g., address planning efficiency)
Documentation	10%	Clear lab reports with topology diagrams and validation steps

5. Tools & Resources

Simulators: GNS3, Cisco Packet Tracer.

Books: *Computer Networks* by Kurose & Ross (IPv6 sections), *IPv6 Essentials* by Silvia Hagen.

Online: AWS IPv6 Addressing Guide , DHCPv6 Tutorial .

Note: Assignments should align with **GATE CSE 2025 syllabus** (Computer Networks: 10% weightage) .

This framework balances theory (address mechanics, protocols) and hands-on practice (configuration, cloud integration), preparing students for real-world IPv6 deployment challenges.